

Nadia Cecil

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SUMMARY

Computer Science undergraduate specializing in Cyber Security with hands-on experience in Python-based backend development, Flask APIs, cybersecurity tools, and AI/ML projects. Built and deployed multiple applications involving web vulnerability analysis, steganography, blockchain concepts, and database-driven systems. Familiar with REST APIs, HTTP workflows, model integration concepts, and currently pursuing Docker and Kubernetes for scalable deployment and infrastructure orchestration.

EDUCATION

Manipal Institute of Technology, Bangalore

B.Tech in Computer Science, Specialization in Cyber Security

2023 – 2027

CGPA: 7.92/10

SKILLS

Programming Languages: Python, Java, SQL, C, JavaScript

Backend & Web Development: Flask, Java Servlets, JSP, REST APIs, HTML, CSS, JavaScript

AI/ML: Machine Learning Fundamentals, OpenCV, TensorFlow (Basics), CNN Concepts, Computational Intelligence

Databases: SQLite, MySQL, JDBC, SQL Queries, Schema Design

Deployment & Tools: Git, GitHub, Apache Tomcat, Render, VS Code, Eclipse IDE

Cloud & DevOps: Docker (Currently Learning), Kubernetes (Currently Learning)

Cybersecurity Tools: Wireshark, Nmap, Suricata IDS

Soft Skills: Problem Solving, Communication, Teamwork, Attention to Detail

Languages: English, Hindi, Kannada, Tamil, French (Basic)

PROJECTS

ScanX Web Vulnerability Scanner | *Python, Flask, HTML/CSS, Requests*

- Developed a Python-based web vulnerability scanner to analyze HTTP security headers and detect common web security misconfigurations.
- Built backend scanning modules using Flask and integrated real-time result rendering through a responsive web interface.
- Implemented request handling, URL validation, and automated security analysis workflows using Python libraries.
- Deployed the application using GitHub and Render to support cloud-hosted accessibility and testing.

StegoSend Secure Steganography Platform | *Python, Flask, OpenCV*

- Built a secure image steganography platform for hiding and retrieving encrypted messages within images.
- Implemented encoding and decoding workflows using Python image-processing libraries and Flask backend APIs.
- Designed secure file upload and message extraction functionality through an interactive frontend interface.
- Explored practical applications of cryptography and secure communication techniques.

Blockchain-Based Secure Transaction System | *Python*

- Implemented a simplified blockchain system demonstrating distributed ledger concepts, block hashing, and chain validation.
- Developed transaction verification workflows and proof-of-integrity mechanisms using Python.
- Explored cryptographic principles including hashing, immutable storage, and secure data verification.

Library Management System | *Java Servlets, JSP, SQLite, JDBC, Apache Tomcat*

- Developed a full-stack web application with admin and member authentication modules.
- Implemented CRUD operations and handled HTTP request-response workflows using Java Servlets.
- Designed relational database schema and executed SQL queries for efficient data management.
- Configured Apache Tomcat deployment and integrated SQLite database connectivity using JDBC.

AI-Powered Food Adulteration Detection System (Ongoing) | *Python, TensorFlow, OpenCV, Flask*

- Developing a machine learning-based food adulteration detection system using image classification techniques.
- Implementing CNN-based models to identify adulteration patterns in food samples through image preprocessing and prediction workflows.
- Integrating Flask APIs for image upload, preprocessing, and model inference.
- Exploring scalable deployment workflows and AI model integration concepts.

RELEVANT COURSEWORK

AI & Machine Learning: Machine Learning Fundamentals, Artificial Intelligence, Computational Intelligence, Data Analytics

Cyber Security: Cryptography, Intrusion Detection Systems, Authentication, Risk Management, Digital Forensics

Systems & Networking: Operating Systems, Computer Networks, HTTP, TCP/IP, Database Management Systems